Aerospace Medicine and Human Performance (AsMA) — in preparation | Original Research — Extended Abstract

TinyDCS: An edge-deployable machine-learning surrogate of the ADRAC altitude-decompression-sickness risk model with continuous-exposure covariates and calibrated uncertainty

Diego Malpica, MD[1*] · Marian Farfán, MD[1]

[1]Subdirectorate of Aerospace Sciences, Direction of Aerospace Medicine, Colombian Aerospace Force, Bogotá DC, Colombia

[*]Correspondence: diego.malpica@fac.mil.co

Running title: TinyDCS: edge surrogate of ADRAC **Word count:** approx. 250 (abstract) — full manuscript in preparation
Version: 0.5.0 — 2026-04-18

ABSTRACT

Background: The US Air Force Altitude DCS Risk Assessment Computer (ADRAC; Pilmanis 2004) is the operational standard for planning hypobaric exposures in aviation and extravehicular activity, but two limitations constrain modern use: its three-level exercise covariate cannot accommodate continuous wearable-derived VO_2 trajectories, and it returns point estimates without calibrated uncertainty.

Objective: To build and benchmark a wearable-grade machine-learning surrogate of the ADRAC grid that (i) accepts continuous-VO_2 exposure covariates, (ii) ships calibrated 95% prediction intervals with uniform altitude-band coverage, (iii) abstains outside the validated input envelope, and (iv) meets an edge-deployment footprint below 100 KB and a per-inference latency below 10 µs.

Methods: We audited and repaired a public ADRAC-output grid (16,295 rows; 1,221 rows, 7.5%, were mis-entered on the fraction scale and were rescaled neighbour-consistently to percent), retaining 15,908 unique grid cells. We trained a LightGBM regressor on the logit of P(DCS) using a 13-feature vector that augments the ADRAC covariates with a Conkin 2004 single-compartment tissue-nitrogen ratio (360-min half-time) and continuous-VO_2 summaries consistent with Webb 2010's 1-minute-peak metric. Smithson–Verkuilen boundary shrinkage handled exact-zero targets, physiological monotonicity constraints were imposed on altitude, prebreathe, time-at-altitude, and tissue-N_2 features, and a two-stage zero-inflated conformal stack was fitted: a binary classifier for the exact-zero mass combined with a split-conformal regressor for the positive support, with Mahalanobis-distance out-of-envelope abstention. We benchmarked against a closed-form log-logistic AFT fit to the same grid and exported the surrogate to ONNX. A conjugate-Gaussian hierarchical Bayesian personalization layer was prototyped on a synthetic 200-subject cohort.

Results: On the held-out random test fold (n = 2,387), TinyDCS attained MAE = 0.020, R^2 = 0.986, and Brier score = 0.0016 — a 4-fold MAE reduction and 10-fold Brier reduction over the closed-form baseline (MAE = 0.086; Brier = 0.0150). Empirical 95% coverage was 0.960 overall and at least 0.95 in each of the five 5,000-ft altitude bands, closing a low-band shortfall (coverage 0.58–0.59 at 18,000–23,000 ft) that was invariant under four conformal-only alternatives and was diagnosed as target-distribution pathology rather than residual-variance error. A compact zero-inflated variant compiled to 95 KB of ONNX with CPU per-row latency of 2.44 µs (p50). The personalization prototype recovered per-subject log-susceptibility at Pearson r

= 0.63 after twenty exposures per subject, with population-vs-personalized Brier parity crossover near $k = 10$.

Conclusions: A continuous-VO_2 ADRAC surrogate with zero-inflated conformal calibration and out-of-envelope abstention outperforms the closed-form model on the same data at an edge-feasible memory and latency budget. External prospective validation on a hypobaric-chamber cohort with Doppler venous-gas-emboli ground truth, and replacement of the conjugate-Gaussian personalization stub with a full hierarchical Bayesian model on real subjects, are the priority follow-ups. The full methods pipeline, trained weights, ONNX exports, and an AI-agent continuation guide (`AGENTS.md`) are released open-source at `github.com/strikerdlm/DCS`.

Keywords: altitude decompression sickness; ADRAC; wearable computing; conformal prediction; zero-inflated models; edge AI; aerospace medicine; hierarchical Bayesian personalization

Repository: github.com/strikerdlm/DCS

Significance

Unpressurized general aviation above FL180, military egress from pressurized cockpits, and EVA pre-breathe scheduling all rely on ADRAC-class closed-form models for ground-side planning. None of the published operational models ingests the per-second VO_2, heart-rate, SpO_2, and barometric-altitude streams that modern wearables now produce. This work delivers a first-generation surrogate that is simultaneously (a) accurate on the ADRAC-grid reference target, (b) honestly uncertainty-quantified across the full altitude range of operational interest, and (c) small and fast enough to run on a flight-watch or smartphone background task — three properties that no prior published altitude-DCS model has achieved together.

Summary of contributions

1. **Data audit.** The shipped `DCS_Risk_DB_2025.csv` contains a systematic scale inconsistency affecting 7.5% of rows; we document the defect and release a deterministic neighbour-median cleaner that restores the grid to 15,908 unique cells on a single scale.
2. **Hybrid feature vector.** A 13-feature descriptor that carries ADRAC's covariates forward while injecting a Conkin 2004 tissue-N_2 ratio and continuous-VO_2 summaries consistent with Webb 2010.
3. **Zero-inflated conformal calibration.** A two-stage calibration that routes the exact-zero mass through a dedicated binary classifier and closes a low-altitude coverage gap that four conformal-only alternatives could not.
4. **Edge-deployable ONNX ladder.** Model-size variants spanning 17 KB to 1.8 MB; the 95 KB compact variant retains $R^2 > 0.98$ and still outperforms the closed-form ADRAC baseline by three-fold on MAE.
5. **Hierarchical personalization prototype.** A conjugate-Gaussian per-subject posterior on log susceptibility, with an explicit information-gain curve across $k = 1$–20 exposures (Paper 2 scope).
6. **Reproducibility package.** A command-by-command runbook, an honest validation-hardware inventory, an NEDU TR 18-01 Appendix-C audit checklist for the open 3RUT-MBe1 reconciliation, and an AI-agent continuation guide.

Data and code availability

All code, cleaned data, trained model bundles (joblib), ONNX artifacts, metrics JSONs, and figure bundles are released under a research-use license at `github.com/strikerdlm/DCS`. Every headline number in this abstract is reproducible end-to-end in under three minutes on CPU via `docs/runbook.md`.

Conflicts of interest and funding

The authors declare no conflicts of interest. This work received no external funding.